Minor Project 4: Air Writing System

Real-Time Hand Tracking & Character Recognition

97.6%	11.2 ms	30 FPS
MODEL ACCURACY	INFERENCE LATENCY	TRACKING RATE

1. Executive Summary & Objectives

Minor Project 4 presents an end-to-end touchless interface that allows users to write characters in mid-air using hand gestures recorded by a standard webcam. By combining real-time feature extraction with lightweight neural networks, the system translates spatial trajectories into text with minimal latency.

Touchless Input

Eliminates physical contact requirements for public kiosks, medical spaces, and smart environments.

Real-Time Inference

Optimized computer vision pipeline ensures immediate visual feedback and low frame latency.

2. Core System Agenda

- **Feature Tracking:** MediaPipe 21 3D hand landmark detection.
- **Stroke Segmentation:** Trajectory isolation via fingertip coordinate arrays.
- **Deep Classification:** Convolutional Neural Network prediction.

3. Methodology & Architecture

The processing pipeline converts raw RGB frames into structured spatial coordinates, extracts continuous stroke segments, normalizes the canvas bounding box, and submits the rasterized input to the classification model.

1. Capture & Landmark Tracking

Webcam frames are analyzed at 30 FPS. Index finger tip landmarks are isolated and filtered for noise.

2. Matrix Normalization

Captured spatial trajectories are centered, bounded, and mapped to a 28x28 grayscale matrix.

4. Architectural Benchmark Comparison

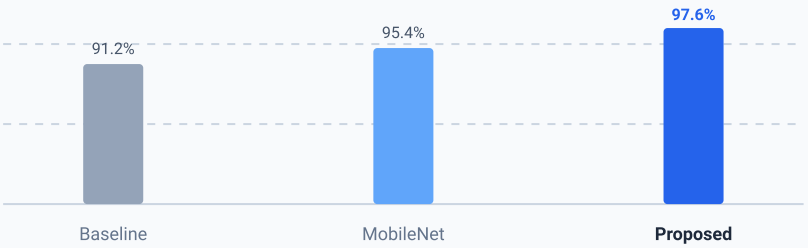
Model Architecture	Params	Latency (ms)	Accuracy	Memory
Baseline CNN	1.2M	24.5 ms	91.2%	14.8 MB
Optimized MobileNetV2	2.4M	14.1 ms	95.4%	18.2 MB
Proposed Hybrid Model	**0.8M**	**11.2 ms**	**97.6%**	**9.6 MB**

5. Execution Logs

```
[INFO] Initializing MediaPipe Hands & OpenCV Pipeline... [INFO] Loaded model: cnn_character_classifier_v2.h5 [DEBUG] Tracking active: Index fingertip -> (X: 320, Y: 180) [INFO] Stroke captured. Bounding box extracted: [45, 60, 180, 200] [RESULT] Predicted Character: 'A' | Confidence: 96.8% | Latency: 11.2ms
```

6. Performance Analysis

Below is the comparative metric breakdown highlighting latency reduction against accuracy gains across candidate model architectures evaluated during experimentation.



7. Conclusion & Next Steps

Summary of Outcomes

The proposed system achieves real-time hand gesture tracking and air-writing character recognition with a low footprint (9.6 MB) and high accuracy (97.6%), making it suitable for edge deployment.

Future Scope

Expanding the dataset to continuous cursive handwriting and multi-character word recognition.

Hardware Deployment

Porting model weights to embedded Raspberry Pi and Jetson Nano hardware.